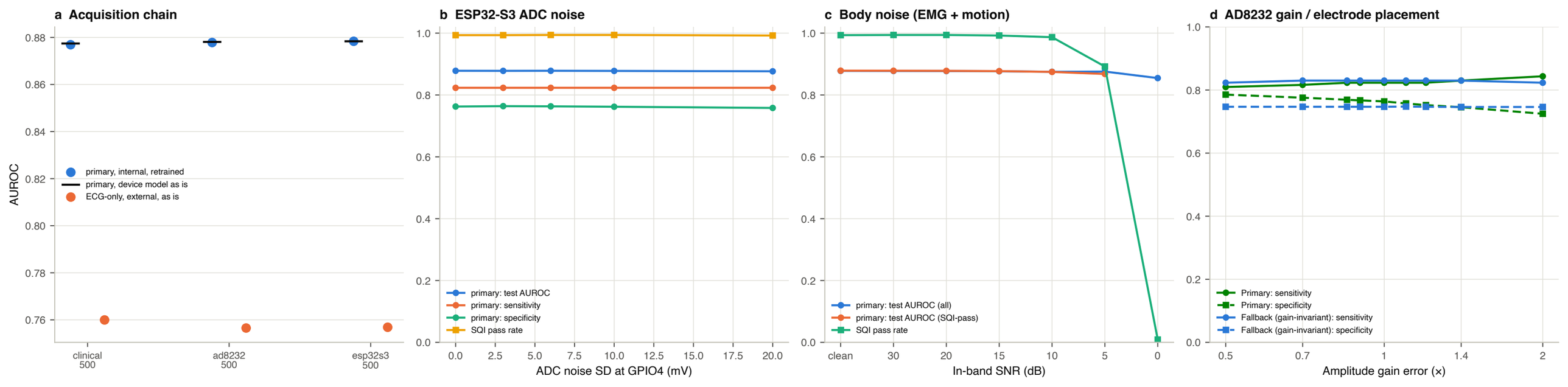
a Acquisition chain
AUROC
0.88
0.86
0.84
0.82
0.80
0.78
0.76
primary, internal, retrained
primary, device model as is
ECG-only, external, as is
clinical
500
ad8232
500
esp32s3
500
b ESP32-S3 ADC noise
1.0
0.8
0.6
0.4
0.2
0.0
primary: test AUROC
primary: sensitivity
primary: specificity
SQI pass rate
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
ADC noise SD at GPIO4 (mV)
c Body noise (EMG + motion)
primary: test AUROC (all)
primary: test AUROC (SQI-pass)
SQI pass rate
clean
30
20
15
10
5
0
In-band SNR (dB)
d AD8232 gain / electrode placement
Primary: sensitivity
Primary: specificity
Fallback (gain-invariant): sensitivity
Fallback (gain-invariant): specificity
0.5
0.7
1
1.4
2
Amplitude gain error (×)